Test case: AL-42

Story: As a borrower, I message my assigned broker or lending officer in the app

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Date run: 6 Nov 2026
Build: `main` at commit `66e32f0`

Preconditions

- Firebase emulators started with the seed data (`firebase emulators:start --project demo-al --import=./seed-data` in `al-core`), al-core and al-ai running locally, the app on the iPhone 17 simulator. Seeded logins use the demo password from `al-core/scripts/seed.js`.
- Signed in as alice@test.com, whose broker is bob@test.com.

Steps

#	Criterion	Steps	Expected result	Automated check	Result
1	1	Open Messages and send "Hello".	One thread with the assigned broker. The message shows straight away and is saved in Firestore.	MessageInboxTests: ASentMessageShowsStraightAwayWhileItSends, OnceFirestoreHasTheMessageItShowsOnce	[x] Pass [ ] Fail
2	2	Reply as bob@test.com from the portal while the app is on another tab, then open Messages.	The reply arrives within 5 seconds; an unread badge shows until the thread is opened.	MessageInboxTests: ListensToTheBorrowersApplicationAndShowsNewMessagesAsTheyArrive, LeavingTheThreadClearsTheBadgeAndTheDivider	[x] Pass [ ] Fail
3	3	Stop al-core and send a message, then start al-core and tap it.	Marked "Not sent: tap to try again"; tapping sends it.	MessageInboxTests.AFailedSendIsMarkedNotSentAndTappingSendsItAgain	[x] Pass [ ] Fail
4	4	Try to read another borrower's thread.	Refused.	Backend rules test (al-core, messaging subtask), plus a check on the emulators	[x] Pass [ ] Fail

Automated test run

iOS unit tests (`AgenticLenderTests`) on `main` at `66e32f0`, iPhone 17 simulator (iOS 26.5), 6 Nov 2026: **651 passed, 0 failed, 0 skipped**. The UI tests named above (`AgenticLenderUITests`) need the emulators running and were not part of this run.

Notes

- Step 1: Messages opened one thread, with Bob Murphy (Your broker · Murphy Mortgages). "Hello Bob" showed at once under Today and was saved in Firestore under `app-applied/messages`.
- Step 2: Home showed a "1 new" badge for the seeded unread reply. A new reply from Bob (sent through POST /core/applications/app-applied/messages as bob@test.com) made the badge show again on Home within 5 seconds; opening the thread showed it under a New messages divider, and back on Home the badge was gone and Firestore had it as read.
- Step 3: with al-core stopped, "Alpha 123" showed greyed with "Not sent. Tap to try again". After al-core was started, tapping it sent it, and Firestore has it once (no duplicate).
- Step 4: signed in as ben@test.com, reading Alice's messages straight from Firestore is refused by the security rules (PERMISSION_DENIED), and al-core answers 404 to his unread count and to sending on her thread.
- The criterion's wording is "Not sent: tap to try again"; the app says "Not sent. Tap to try again".